HEART DISEASE PREDICTION USING MACHINE LEARNING

Internship Project Report

Submitted By

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Internship Program

CodeAlpha – Machine Learning Internship

Project Title

Heart Disease Prediction Using Machine Learning

Abstract

Heart disease is one of the leading causes of mortality worldwide. Early detection of heart disease can assist healthcare professionals in providing timely treatment and improving patient outcomes. This project focuses on developing a machine learning-based system capable of predicting the presence of heart disease using patient medical data.

Various machine learning algorithms were implemented and evaluated to identify the most effective predictive model. The project demonstrates how data-driven approaches can support medical diagnosis and decision-making.

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1. Introduction

Heart disease affects millions of people globally and remains a major public health concern. Traditional diagnostic procedures often require extensive medical examinations and expert interpretation.

Machine learning offers a powerful solution by identifying hidden patterns in patient data and providing accurate predictions. The goal of this project is to build a predictive model that can determine whether a patient is likely to have heart disease based on clinical attributes.

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2. Objectives

The main objectives of this project are:

- To analyze heart disease patient data.
- To preprocess and clean medical records.
- To train machine learning models for disease prediction.
- To compare model performance using evaluation metrics.
- To identify the most accurate model for heart disease prediction.

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3. Dataset Description

The project utilizes a heart disease dataset containing patient medical information.

Dataset Characteristics

- Total Records: 918
- Target Variable: Heart Disease Presence
- Type: Structured Medical Dataset

Features Included

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level
- Fasting Blood Sugar
- Resting ECG Results
- Maximum Heart Rate
- Exercise-Induced Angina
- Old Peak
- ST Slope
- Number of Major Vessels
- Thalassemia
- Target (Heart Disease)

These features are commonly used by healthcare professionals to assess cardiovascular risk.

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4. Methodology

4.1 Data Collection

The heart disease dataset was loaded into Python for analysis and model development.

4.2 Data Preprocessing

The following preprocessing steps were performed:

- Inspection of dataset structure
- Statistical analysis using descriptive statistics
- Identification of missing values
- Data cleaning
- Feature preparation for machine learning

4.3 Exploratory Data Analysis

Exploratory analysis was performed to understand:

- Distribution of patient attributes
- Feature statistics
- Relationships among variables
- Disease occurrence patterns

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5. Machine Learning Models Used

Three supervised learning algorithms were implemented and evaluated.

5.1 Logistic Regression

Logistic Regression is a statistical classification algorithm used for binary prediction problems. It estimates the probability of disease occurrence using a logistic function.

Advantages

- Simple and interpretable
- Fast training time
- Effective baseline model

5.2 Support Vector Machine (SVM)

Support Vector Machine is a powerful classification technique that finds the optimal decision boundary between classes.

Advantages

- High classification accuracy
- Effective in high-dimensional datasets
- Good generalization capability

5.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance.

Advantages

- Handles complex relationships
- Reduces overfitting
- Robust to noise

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6. Model Evaluation Metrics

The following metrics were used:

Accuracy

Measures the percentage of correct predictions.

Precision

Measures the proportion of positive predictions that are actually correct.

Recall

Measures the proportion of actual positive cases correctly identified.

F1-Score

Harmonic mean of Precision and Recall.

ROC-AUC Score

Measures the model's ability to distinguish between classes.

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7. Results

Logistic Regression

- Accuracy: 76.09%
- ROC-AUC Score: 0.8371

Support Vector Machine (SVM)

- Accuracy: 78.26%
- ROC-AUC Score: 0.8409

Random Forest

- Accuracy: 76.09%
- ROC-AUC Score: 0.8229

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8. Performance Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	76.09%	0.8371
SVM	78.26%	0.8409
Random Forest	76.09%	0.8229

Best Performing Model

Support Vector Machine (SVM) achieved the highest accuracy and ROC-AUC score, making it the best-performing model in this project.

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9. Conclusion

This project successfully developed a machine learning-based heart disease prediction system using patient medical data. Multiple classification algorithms were implemented and compared.

The experimental results showed that the Support Vector Machine model provided the best predictive performance with an accuracy of approximately 78.26%. The findings demonstrate the effectiveness of machine learning techniques in assisting healthcare professionals with early disease detection and risk assessment.

Such predictive systems can support clinical decision-making and contribute to improved patient care.

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10. Future Scope

Future improvements may include:

- Hyperparameter optimization
- Feature selection techniques
- Deep learning approaches
- Integration with healthcare applications
- Real-time disease prediction systems
- Deployment as a web application using Flask or Streamlit

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

References

1. Scikit-learn Documentation
2. UCI Heart Disease Dataset
3. Machine Learning for Healthcare Research Papers
4. Python Data Science Libraries Documentation

Submitted as part of the CodeAlpha Machine Learning Internship.